Lab 6

Deepshikha Chowdhury UG/02/BTCSEAIML/2023/063

```
test> use college_UG_02_BTCSEAIML_2023_063
switched to db college_UG_02_BTCSEAIML_2023_063
college_UG_02_BTCSEAIML_2023_063> db.createCollection("student_UG_02_BTCSEAIML_2023_063")
{ ok: 1 }
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.insertMany([
... { emp_id: 101, name: "Amit", department: "IT", designation: "Developer", city: "Kolkata", salary: 60000, experience: 3 },
... { emp_id: 102, name: "Neha", department: "HR", designation: "HR Executive", city: "Delhi", salary: 45000, experience: 2 },
... { emp_id: 103, name: "Ravi", department: "IT", designation: "Tester", city: "Mumbai", salary: 50000, experience: 4 },
... { emp_id: 104, name: "Pooja", department: "Finance", designation: "Accountant", city: "Chennai", salary: 55000, experience: 5 },
... { emp_id: 105, name: "Karan", department: "IT", designation: "Developer", city: "Kolkata", salary: 65000, experience: 6 },
... { emp_id: 106, name: "Sneha", department: "HR", designation: "Manager", city: "Delhi", salary: 70000, experience: 7 },
... { emp_id: 107, name: "Arjun", department: "Finance", designation: "Analyst", city: "Mumbai", salary: 48000, experience: 2 },
... { emp_id: 108, name: "Meera", department: "IT", designation: "Developer", city: "Bangalore", salary: 75000, experience: 8 },
... { emp_id: 109, name: "Rohit", department: "IT", designation: "Team Lead", city: "Pune", salary: 90000, experience: 10 },
... { emp_id: 110, name: "Anjali", department: "HR", designation: "Recruiter", city: "Kolkata", salary: 42000, experience: 1 }
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69b788d7346284aa08628ca0'),
    '1': ObjectId('69b788d7346284aa08628ca1'),
    '2': ObjectId('69b788d7346284aa08628ca2'),
    '3': ObjectId('69b788d7346284aa08628ca3'),
    '4': ObjectId('69b788d7346284aa08628ca4'),
    '5': ObjectId('69b788d7346284aa08628ca5'),
    '6': ObjectId('69b788d7346284aa08628ca6'),
    '7': ObjectId('69b788d7346284aa08628ca7'),
    '8': ObjectId('69b788d7346284aa08628ca8'),
    '9': ObjectId('69b788d7346284aa08628ca9')
  }
}
```

Q1. Display employee name and salary only.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find(
... {},
... {name:1,salary:1,_id:0}
... )
[
  { name: 'Amit', salary: 60000 },
  { name: 'Neha', salary: 45000 },
  { name: 'Ravi', salary: 50000 },
  { name: 'Pooja', salary: 55000 },
  { name: 'Karan', salary: 65000 },
  { name: 'Sneha', salary: 70000 },
  { name: 'Arjun', salary: 48000 },
  { name: 'Meera', salary: 75000 },
  { name: 'Rohit', salary: 90000 },
  { name: 'Anjali', salary: 42000 }
]
```

Q2. Find employees working in HR department.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find({department:"HR"})
[
  {
    _id: ObjectId('69b788d7346284aa08628ca1'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 45000,
    experience: 2
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca5'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca9'),
    emp_id: 110,
    name: 'Anjali',
    department: 'HR',
    designation: 'Recruiter',
    city: 'Kolkata',
    salary: 42000,
    experience: 1
  }
]
```

Q3. Find employees whose salary is between 50,000 and 80,000.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find(
... {salary:{$gte:50000,$lte:80000}}
... )
[
  {
    _id: ObjectId('69b788d7346284aa08628ca0'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca2'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca3'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
```

Q4. Display employees from Kolkata city.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find({city:"Kolkata"})
[
  {
    _id: ObjectId('69b788d7346284aa08628ca0'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca4'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca9'),
    emp_id: 110,
    name: 'Anjali',
    department: 'HR',
    designation: 'Recruiter',
    city: 'Kolkata',
    salary: 42000,
    experience: 1
  }
]
```

Q5. Sort employees by experience in ascending order.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find().sort({experience:1})
[
  {
    _id: ObjectId('69b788d7346284aa08628ca9'),
    emp_id: 110,
    name: 'Anjali',
    department: 'HR',
    designation: 'Recruiter',
    city: 'Kolkata',
    salary: 42000,
    experience: 1
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca1'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 45000,
    experience: 2
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca6'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca0'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
```

Q6. Display top 5 most experienced employees.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find()
... .sort({experience:-1})
... .limit(5)
[
  {
    _id: ObjectId('69b788d7346284aa08628ca8'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca7'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca5'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca4'),
    emp_id: 105,
```

Q7. Update salary of employee with emp_id 102 to 48,000.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.updateOne(
... {emp_id:102},
... {$set:{salary:48000}}
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Q8. Increase salary of all IT employees by 10,000.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.updateMany(
... {department:"IT"},
... {$inc:{salary:10000}}
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
```

Q9. Update designation to “Senior Developer” for IT employees having experience more than 5 years.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.updateMany(
... {
... department:"IT",
... experience:{$gt:5}
... },
... {$set:{designation:"Senior Developer"}}
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
```

Q10. Delete employees whose experience is less than 2 years.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.deleteMany(
... {experience:{$lt:2}}
... )
{ acknowledged: true, deletedCount: 1 }
```

Q11. Display employees who are not from Delhi.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find(
... {city:{$ne:"Delhi"}}
... )
[
  {
    _id: ObjectId('69b788d7346284aa08628ca0'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 70000,
    experience: 3
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca2'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 60000,
    experience: 4
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca3'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
```

Q12. Find employees whose designation is Developer and salary greater than 60,000.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find(
... {
... designation:"Developer",
... salary:{$gt:60000}
... }
... )
[
  {
    _id: ObjectId('69b788d7346284aa08628ca0'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 70000,
    experience: 3
  }
]
```

Q13. Display employees working in IT or Finance department.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find(
... {
... $or:[
... {department:"IT"},
... {department:"Finance"}
... ]
... }
... )
[
  {
    _id: ObjectId('69b788d7346284aa08628ca3'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca6'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca0'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 70000,
    experience: 3
```

Q14. Display employees sorted by department and then by salary.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find()
... .sort({department:1,salary:1})
[
  {
    _id: ObjectId('69b788d7346284aa08628ca6'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca3'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca1'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca5'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca2'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
```

Q15. Count total number of employees in the company.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.countDocuments()
9
```

Q16. Display employees whose salary is not equal to 50,000.

```
college_UG_02_BTCSEAIML_2023_063> db.student_UG_02_BTCSEAIML_2023_063.find(
... {salary:{$ne:50000}}
... )
[
  {
    _id: ObjectId('69b788d7346284aa08628ca0'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 70000,
    experience: 3
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca1'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca2'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 60000,
    experience: 4
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca3'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('69b788d7346284aa08628ca4'),
    emp_id: 105,
    name: 'Karan',
```